Kaleem Nawaz Khan

Webpage: https://kaleemnwzkhan.github.io/ Email: kk5271@rit.edu
Interests: Systems + AI, Computer and Mobile Networks, Autonomous Vehicle Systems Mobile: +1-929-832-6936

Research & Technical Profile

My research focuses on perception for autonomous mobile systems. I specialize in cooperative perception, multi-sensor fusion, localization, and 3D scene representation. I develop systems that combine sensing and computing across vehicles and infrastructure. These systems improve the accuracy and coverage of environmental perception for real-time decision-making and safer autonomous operation. My work has appeared in top mobile and sensing systems venues, including *MobiSys* and *SenSys*.

Education

- **Rochester Institute of Technology (RIT)** *Aug. 2022 – Present*
  Ph.D. in Computing and Information Sciences, CGPA: 4.00/4.00 Rochester, NY
- **FAST-National University of Computer and Emerging Sciences** *Aug. 2015 – Dec. 2017*
  M.S. in Computer Science, CGPA: 3.68/4.00 Peshawar, Pakistan
- **FAST-National University of Computer and Emerging Sciences** *Sep. 2011 – Jun. 2015*
  B.S. in Computer Science, CGPA: 3.71/4.00 Peshawar, Pakistan

Work Experience

- **Graduate Research Assistant** *Aug. 2022 – Present*
  Networked Sensing Systems Lab (NSSL), RIT Rochester, NY
  – Developed cooperative perception systems for connected and autonomous vehicles.
  – Co-designed perception, communication, and computing frameworks for reliable autonomous operation.
- **Research Intern** *May. 2024 – Aug. 2024*
  Connected Vehicle Experience Research Lab, General Motors Warren, MI
  – Designed intelligent data offloading mechanisms for connected vehicles leveraging edge and cloud resources.
  – Modeled and evaluated vehicle-to-edge communication using OMNeT++ and Simu5G across network scenarios.
- **Research Team Lead** *Apr. 2019 – Sep. 2020*
  National Center of Artificial Intelligence (NCAI) Peshawar, PK
  – Led a diverse research team to develop AI solutions for healthcare applications.
  – Applied deep learning techniques for medical screening to assist cardiology and hematology practitioners.

Research Projects

- **FRC: Radar–Camera Fusion for Intersection Awareness** *Nov. 2025 – Present*
  Research Assistant – Networked Sensing Systems Lab *Skills: Foundation Models, VLMs, Sensors Fusion*
  – A multimodal perception framework that fuses automotive radar and camera data for intersection awareness.
  – Spatial alignment and synchronization techniques to associate radar detections with camera observations.
- **PMC: Collaborative Intersection Perception Using Multi-Vehicle Cameras** *May 2025 – Present*
  Research Assistant – Networked Sensing Systems Lab *Skills: Reconstruction, SfM, Gaussian Splatting (3DGS)*
  – A collaborative perception system that reconstructs and updates 3D intersection models using vehicle cameras.
  – Leveraged object segmentation and structure-from-motion techniques to estimate dynamic objects in real time.
- **ARC: Accurate, Real-Time, and Scalable Multi-Vehicle Cooperative Perception** *Jun. 2024 – Mar. 2025*
  Research Assistant – Networked Sensing Systems Lab *Skills: LiDAR, Point Cloud, Spatial Alignment, GPU*
  – Designed a scalable cooperative perception framework with high accuracy across multiple vehicles.
  – Developed an alignment refinement layer to improve point cloud registration accuracy in cooperative perception.
  – GPU-accelerated grid-based spatial reasoning to reduce alignment latency and enable selective inter-vehicle sharing.
  – Resulted in a publication at ACM/IEEE SenSys 2026.
- **Smart Vehicle-to-Edge Computation and Data Offloading** *May. 2024 – Aug. 2024*
  Research Intern – Connected Vehicle Experience Research Lab *Skills: Simu5G/OMNeT++, Machine Learning*
  – Developed a framework for intelligent vehicular computation and data offloading to edge and cloud infrastructure.
  – Simulated 5G vehicular networks in Simu5G/OMNeT++ with mobile vehicles, base stations, and edge servers.
  – Evaluated effects of latency, signal quality, and congestion on offloading performance under varying traffic conditions.

- **VRF: Vehicle Road-side Point Cloud Fusion** *Sep. 2022 – Mar. 2024*
  Research Assistant – Networked Sensing Systems Lab *Skills: LiDAR, Accuracy Forecasting, 3D Mapping*
  - Cooperative perception framework for real-time fusion of vehicle and roadside LiDAR data to mitigate occlusions.
  - Map-based alignment strategy to independently align vehicle and roadside point clouds to a shared reference frame.
  - Demonstrated real-world performance with 20 ms latency and 5 cm positioning accuracy on testbeds.
  - Resulted in a publication at ACM MobiSys 2024.

Publications

- [**SenSys'26**] ARC: Accurate, Real-Time, and Scalable Multi-Vehicle Cooperative Perception
  K. N. Khan, F. Ahmad
  Proceedings of the 2026 ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems
- [**MobiSys'24**] VRF: Vehicle Road-side Point Cloud Fusion
  K. N. Khan, A. Khalid, Y. Turkar, K. Dantu, F. Ahmad
  Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications, and Services, 2024
- [**SN'22**] Op2Vec: An Opcode Embedding Technique and Dataset Design for End-to-End Detection of Android Malware
  K. N. Khan, N. Ullah, S. Ali, M. S. Khan, M. Nauman, A. Ghani
  Security and Communication Networks, 2022
- [**Diagnostics'22**] Deep Learning Assisted Automated Assessment of Thalassaemia from Hb Electrophoresis Images
  M. S. Khan, A. Ullah, **K. N. Khan**, H. Riaz, Y. M. Yousafzai, T. Rahman, M. E. Chowdhury, S. B. Abul Kashem
  Diagnostics, Vol. 12, No. 10, 2022
- [**Physiol. M.'21**] Deep Learning Based Classification of Unsegmented PCG Spectrograms Leveraging Transfer Learning
  K. N. Khan, F. A. Khan, A. Abid, T. Olmez, Z. Dokur, A. Khandakar, M. E. Chowdhury, M. S. Khan
  Physiological Measurement, Vol. 42, No. 9, 2021
- [**ICAI'21**] Vision-based Human Posture Classification and Fall Detection Using Convolutional Neural Networks
  R. Hasib, **K. N. Khan**, M. Yu, M. S. Khan
  IEEE 2021 International Conference on Artificial Intelligence (ICAI)
- [**ICAI'21**] Performance Evaluation of Background Subtraction Techniques for Video Frames
  S. Qasim, **K. N. Khan**, M. Yu, M. S. Khan
  IEEE 2021 International Conference on Artificial Intelligence (ICAI)
- [**ICOSST'21**] Automatic Classification of Heart Sounds Using Long Short-Term Memory
  B. Ahmad, F. A. Khan, **K. N. Khan**, M. S. Khan
  IEEE 2021 15th International Conference on Open Source Systems and Technologies (ICOSST)

Awards, Grants, and Fellowships

- **Travel Grant**, Rochester Institute of Technology (RIT), to present research at ACM MobiSys 2024, Tokyo, Japan.
- **M.S. Fellowship**, FAST-National University, Peshawar, Pakistan (2015–2017).
- **Prime Minister National ICT R&D Fund Scholarship** for B.S. in Computer Science, Pakistan (2011–2015).
- **Silver Medal**, B.S. in Computer Science, FAST-National University, Peshawar, Pakistan (2011–2015).
- **Stoori da Pakhtunkhwa Merit Certificate**, Government of Khyber Pakhtunkhwa, Pakistan.

Academic Services and Memberships

- **Conference Reviewer:** Shadow Program Committee of EuroSys 2026, MobiSys'26 Artifact Evaluation Committee.
- **Membership:** ACM-Student Member, RIT-Doctoral Students Association, RIT-Pak Students Association.
- **Volunteer**, Imagine RIT 2026, contributed to event operations at a campus-wide creativity and innovation festival.

Presentations and Talks

- **Conference Talk**, ACM MobiSys'24, International Conference on Mobile Systems, Applications, and Services, Tokyo
- **Research Showcase**, Presented work on cooperative perception at Imagine RIT, RIT's annual innovation festival.

Skills Summary

Programming & Software: Python, C++, C, SQL, ROS, Docker, CUDA, LaTeX
AI & Computer Vision: Machine Learning, Transformers, 3D Perception, Multi-Sensor Fusion, Scene Understanding
Sensing & Localization: LiDAR, Stereo Vision, Point Clouds, SLAM, Localization
3D Reconstruction: NeRF, 3D Gaussian Splatting (3DGS), Structure-from-Motion (SfM)
Simulation: CARLA, OMNeT++